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## 1 The infinite spine case

Throughout this section,  $G$  denotes an ordered abelian group. For  $n \geq 2$  and  $a \notin nG$ , recall that  $\mathfrak{s}_n(a)$  is the largest convex subgroup  $H \subseteq G$  such that  $a \notin H + nG$ . If  $a \in nG$ , we set  $\mathfrak{s}_n(a) = \emptyset$ , and we regard  $\emptyset$  as being smaller than every convex subgroup. We set  $\mathcal{S}_n = \{\mathfrak{s}_n(a) \mid a \in G\}$ . The subgroups  $\mathfrak{s}_n(a)$  are uniformly definable in  $a$  [1, Lemma 2.1]. Consequently, comparisons and equalities between values of the maps  $\mathfrak{s}_n$  are formulas in  $L_{\text{oag}}$ .

**Lemma 1.1** ([3, Lemma 2.8(2) and Remark 2.6]). *Let  $p$  be a prime,  $r \geq 1$ , and  $a, b \in G$ .*

1. *If  $0 \leq j < r$ , then*

$$\mathfrak{s}_{p^r}(p^j a) = \mathfrak{s}_{p^{r-j}}(a).$$

*If  $j \geq r$ , then  $\mathfrak{s}_{p^r}(p^j a) = \emptyset$ .*

2. *If  $\mathfrak{s}_{p^r}(a) \neq \mathfrak{s}_{p^r}(b)$ , then*

$$\mathfrak{s}_{p^r}(a + b) = \max\{\mathfrak{s}_{p^r}(a), \mathfrak{s}_{p^r}(b)\}.$$

**Lemma 1.2.** *Let  $p$  be a prime and let  $H < H'$  be nonempty elements of  $\mathcal{S}_p$ . There exists  $a \in H'$  such that*

$$\mathfrak{s}_p(a) = H \quad \text{and} \quad \mathfrak{s}_{p^r}(a) < H'$$

*for every  $r \geq 1$ .*

*Proof.* Choose  $b \in G$  such that  $\mathfrak{s}_p(b) = H$ . Since  $H' > H$ , the maximality of  $H$  gives  $b \in H' + pG$ . We may therefore write  $b = a + pc$  with  $a \in H'$  and  $c \in G$ . As  $a - b \in pG$ , we have  $\mathfrak{s}_p(a) = \mathfrak{s}_p(b) = H$ .

Fix  $r \geq 1$ . If  $\mathfrak{s}_{p^r}(a) = \emptyset$ , there is nothing to prove. Otherwise, put  $K = \mathfrak{s}_{p^r}(a)$ . If  $K \geq H'$ , then  $a \in H' \subseteq K$ , contrary to  $a \notin K + p^r G$ . Hence,  $K < H'$ . ■

**Lemma 1.3** (Finite coding). *Let  $p$  be a prime and*

$$K_0 < K_1 < \cdots < K_N$$

*be nonempty elements of  $\mathcal{S}_p$ . There exists  $x \in G$  such that*

$$\mathfrak{s}_{p^{i+1}}(x) = K_i$$

*for every  $0 \leq i \leq N$ .*

*Proof.* For each  $j < N$ , use Lemma 1.2 to choose  $a_j \in G$  such that

$$\mathfrak{s}_p(a_j) = K_j \quad \text{and} \quad \mathfrak{s}_{p^r}(a_j) < K_{j+1}$$

for every  $r \geq 1$ . Also choose  $a_N \in G$  with  $\mathfrak{s}_p(a_N) = K_N$ , and set

$$x = a_0 + pa_1 + \cdots + p^N a_N.$$

Fix  $0 \leq i \leq N$ . By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i) = \mathfrak{s}_p(a_i) = K_i.$$

If  $j < i$ , then

$$\mathfrak{s}_{p^{i+1}}(p^j a_j) = \mathfrak{s}_{p^{i+1-j}}(a_j) < K_{j+1} \leq K_i,$$

while, if  $j > i$ , then  $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \emptyset$ . Thus, among the summands of  $x$ , the unique maximal  $\mathfrak{s}_{p^{i+1}}$ -value is  $K_i$ . Repeatedly applying Lemma 1.1(2) gives  $\mathfrak{s}_{p^{i+1}}(x) = K_i$ .  $\blacksquare$

**Theorem 1.4.** *If  $\mathcal{S}_p$  is infinite for some prime  $p$ , then the  $dp$ -rank of  $G$  is  $\aleph_0$ .*

*Proof.* We work in a sufficiently saturated monster model of  $\text{Th}(G)$ . Since  $\mathcal{S}_p$  is an infinite interpretable linear order, compactness allows us to choose elements

$$(b_{i,k})_{i,k < \omega}$$

such that all the values

$$H_{i,k} := \mathfrak{s}_p(b_{i,k})$$

are nonempty and

$$H_{i,k} < H_{j,l} \quad \text{whenever} \quad i < j \text{ or } (i = j \text{ and } k < l).$$

Indeed, every finite part of these requirements merely asks for a finite chain of distinct nonempty elements of the infinite linear order  $\mathcal{S}_p$ .

For a function  $\eta : \omega \rightarrow \omega$ , consider the partial type

$$\Sigma_\eta(x) = \{ \mathfrak{s}_{p^{i+1}}(x) = H_{i,\eta(i)} \mid i < \omega \}.$$

This type is finitely satisfiable. Indeed, given  $N < \omega$ , the values

$$H_{0,\eta(0)} < H_{1,\eta(1)} < \cdots < H_{N,\eta(N)}$$

form a finite increasing chain, so Lemma 1.3 realizes the first  $N + 1$  conditions of  $\Sigma_\eta$ . By saturation, there is  $a_\eta \in G$  realizing  $\Sigma_\eta$ .

For every  $i < \omega$ , let

$$\varphi_i(x; y) \text{ be the formula } \mathfrak{s}_{p^{i+1}}(x) = \mathfrak{s}_p(y).$$

As noted above, this is an  $L_{\text{oag}}$ -formula. For all  $\eta : \omega \rightarrow \omega$ ,  $i < \omega$ , and  $k < \omega$ , we have

$$\models \varphi_i(a_\eta; b_{i,k}) \iff H_{i,\eta(i)} = H_{i,k} \iff \eta(i) = k.$$

Hence, these formulas and parameters form an ICT-pattern of depth  $\aleph_0$ , and therefore the dp-rank of  $G$  is at least  $\aleph_0$ .

Every ordered abelian group has NIP [2]. Since  $L_{\text{oag}}$  is countable, this gives dp-rank at most  $\aleph_0$ . Consequently, the dp-rank of  $G$  is exactly  $\aleph_0$ .  $\blacksquare$

## 2 The lower bound in the finite-spine case

We now assume that  $G$  is nontrivial and has finite spines. For a prime  $p$  and  $H \in \mathcal{S}_p \setminus \{\emptyset\}$ , recall that

$$H^{[p]} = \bigcap_{K \in G, K > H} (K + pG)$$

and

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

by [3, Definition 2.10 and Lemma 2.14(1)]. We also use  $\mathcal{S}_{p^r} = \mathcal{S}_p$  for every  $r \geq 1$ , as established in [3, Lemma 2.9(3)]. We put

$$\mathcal{S}_p^\infty = \{H \in \mathcal{S}_p \setminus \{\emptyset\} \mid |R_H^{\mathfrak{s}_p}| = \infty\}.$$

**Lemma 2.1.** *Let  $p$  be a prime and  $H \in \mathcal{S}_p^\infty$ . There is a sequence  $(a^k)_{k < \omega}$  such that*

$$\mathfrak{s}_p(a^k) = \mathfrak{s}_p(a^k - a^l) = H$$

*whenever  $k \neq l$ . If  $H < K$  for some  $K \in \mathcal{S}_p \setminus \{\emptyset\}$ , the sequence may moreover be chosen inside  $K$ .*

*Proof.* Choose elements  $(u^k)_{k < \omega}$  whose images in

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

are nonzero and pairwise distinct. Thus, neither  $u^k$  nor  $u^k - u^l$  belongs to  $H + pG$ , while all these elements belong to  $H^{[p]}$ . By [3, Lemma 2.14(1)], this says precisely that

$$\mathfrak{s}_p(u^k) = \mathfrak{s}_p(u^k - u^l) = H$$

for  $k \neq l$ .

Suppose that  $H < K$ . Since  $u^k \in H^{[p]} \subseteq K + pG$ , we can write  $u^k = a^k + pc_k$  with  $a^k \in K$ . Replacing  $u^k$  by  $a^k$  does not change its image in  $R_H^{\mathfrak{s}_p}$ . The images therefore remain nonzero and pairwise distinct, so the required equalities remain true.  $\blacksquare$

**Lemma 2.2** (Simultaneous coding for one prime). *Let  $p$  be a prime and let*

$$H_0 < H_1 < \cdots < H_{n-1}$$

*be elements of  $\mathcal{S}_p^\infty$ . For each  $i < n$ , there is a sequence  $(a_i^k)_{k < \omega}$  as in Lemma 2.1, and these sequences can be chosen so that  $a_i^k \in H_{i+1}$  whenever  $i < n-1$ . For  $\zeta : n \rightarrow \omega$ , set*

$$b_\zeta = a_0^{\zeta(0)} + pa_1^{\zeta(1)} + \cdots + p^{n-1}a_{n-1}^{\zeta(n-1)}.$$

*Then, for every  $i < n$ ,*

1.  $\mathfrak{s}_{p^{i+1}}(b_\zeta) = H_i$ ;
2. the image of  $b_\zeta$  in  $R_{H_i}^{\mathfrak{s}_{p^{i+1}}}$  is the same as that of  $p^i a_i^{\zeta(i)}$ ;
3. for every  $k < \omega$ ,

$$b_\zeta - p^i a_i^k \in H_i + p^{i+1}G \iff k = \zeta(i).$$

*Proof.* The required sequences are supplied by Lemma 2.1. For  $j < n-1$ , the fact that  $a_j^k \in H_{j+1}$  implies

$$\mathfrak{s}_{p^r}(a_j^k) < H_{j+1}$$

for every  $r \geq 1$ : otherwise,  $\mathfrak{s}_{p^r}(a_j^k) \geq H_{j+1}$  would contain  $a_j^k$ , contrary to the definition of  $\mathfrak{s}_{p^r}$ . The calculation in the proof of Lemma 1.3 now gives

$$\mathfrak{s}_{p^{i+1}}(b_\zeta) = H_i,$$

which proves (1).

For  $j < i$ , we have  $a_j^{\zeta(j)} \in H_{j+1} \subseteq H_i$ , whereas for  $j > i$ , the term  $p^j a_j^{\zeta(j)}$  belongs to  $p^{i+1}G$ . Hence,

$$b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G.$$

Also, by Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i^{\zeta(i)}) = \mathfrak{s}_p(a_i^{\zeta(i)}) = H_i.$$

Thus, both elements represent classes in

$$R_{H_i}^{\mathfrak{s}_{p^{i+1}}} = H_i^{[p^{i+1}]} / (H_i + p^{i+1}G),$$

and the congruence  $b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G$  proves (2).

If  $k \neq \zeta(i)$ , then

$$p^i (a_i^{\zeta(i)} - a_i^k) \notin H_i + p^{i+1}G.$$

Indeed, by the purity of the convex subgroup  $H_i$ , membership in the group on the right would imply

$$a_i^{\zeta(i)} - a_i^k \in H_i + pG,$$

contrary to Lemma 2.1. Together with the preceding congruence, this proves (3).  $\blacksquare$

The following standard consequences of compactness will be used without proof. For the second assertion, finite satisfiability follows from the Chinese remainder theorem [3, Lemma 2.1(2)].

**Lemma 2.3.** *Let  $G$  be a sufficiently saturated monster model of the theory of a nontrivial ordered abelian group.*

1. *The type-definable subgroup*

$$D = \bigcap_{m \geq 2} mG$$

*is cofinal and coinital in  $G$ .*

2. *Let  $P_0 \subseteq \mathbb{P}$  and  $(a_p)_{p \in P_0}$  be a sequence from  $G$ . There exists  $a \in G$  such that*

$$a - a_p \in p^r G$$

*for every  $p \in P_0$  and  $r \geq 1$ .*

**Theorem 2.4.** *Let  $G$  be a nontrivial ordered abelian group with finite spines. Then*

$$\text{dp-rk}(G) \geq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

*If the sum on the right is infinite, then the dp-rank of  $G$  is  $\aleph_0$ .*

*Proof.* We work in a sufficiently saturated monster model. For each prime  $p$ , put  $n_p = |\mathcal{S}_p^\infty|$  and, when  $n_p > 0$ , enumerate

$$\mathcal{S}_p^\infty = \{H_{p,0} < \cdots < H_{p,n_p-1}\}.$$

The number  $n_p$  is finite because  $G$  has finite spines. Let

$$\mathcal{I} = \{(p, i) \mid p \in \mathbb{P}, 0 \leq i < n_p\}.$$

Thus,

$$|\mathcal{I}| = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

and  $\mathcal{I}$  is finite or countable.

For each  $p$  with  $n_p > 0$ , choose sequences  $(a_{p,i}^k)_{k < \omega}$  as in Lemma 2.2, and denote the corresponding coding elements by  $b_{p,\zeta}$  for  $\zeta : n_p \rightarrow \omega$ . For  $(p, i) \in \mathcal{I}$ , let

$$\varphi_{p,i}(x; y) \text{ be the formula } x - y \in \mathfrak{s}_{p^{i+1}}(y) + p^{i+1}G.$$

This is an  $L_{\text{oag}}$ -formula by the uniform definability of  $\mathfrak{s}_{p^{i+1}}$ . We use the parameter sequence

$$(p^i a_{p,i}^k)_{k < \omega}$$

in the row indexed by  $(p, i)$ . By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_{p,i}^k) = H_{p,i}.$$

Fix a function  $\eta : \mathcal{I} \rightarrow \omega$  and consider

$$q_\eta(x) = \left\{ \varphi_{p,i}(x; p^i a_{p,i}^k)^{\eta(p,i)=k} \mid (p,i) \in \mathcal{I}, k < \omega \right\}.$$

Here, the superscript means that the formula is negated when  $\eta(p,i) \neq k$ . For each  $p$  with  $n_p > 0$ , let

$$\eta_p : n_p \rightarrow \omega, \quad \eta_p(i) = \eta(p,i).$$

By Lemma 2.3(2), there is  $b_\eta \in G$  such that

$$b_\eta - b_{p,\eta_p} \in p^r G$$

for every  $p$  with  $n_p > 0$  and  $r \geq 1$ . Part (3) of Lemma 2.2 then shows that  $b_\eta \models q_\eta$ .

It remains to add the row arising from the order. Since the monster is sufficiently saturated, there exists  $c > 0$  such that

$$-c < b_\eta < c$$

for every  $\eta : \mathcal{I} \rightarrow \omega$ . By Lemma 2.3, choose

$$t_0 < t_1 < \dots$$

from  $D$  so that  $t_{k+1} - t_k > 2c$  for every  $k < \omega$ . Set

$$I_k = (t_k - c, t_k + c).$$

These intervals are pairwise disjoint and satisfy  $I_0 < I_1 < \dots$ . We take

$$\psi(x; y, z) := (y < x < z)$$

for the additional row, with parameter sequence

$$(t_k - c, t_k + c)_{k < \omega}.$$

Let  $\nu : \{*\} \dot{\cup} \mathcal{I} \rightarrow \omega$ , and put  $\eta = \nu|_{\mathcal{I}}$ . The element

$$a_\nu = b_\eta + t_{\nu(*)}$$

belongs to  $I_{\nu(*)}$ . Moreover,  $t_{\nu(*)} \in p^{i+1}G$  for every  $(p,i) \in \mathcal{I}$ , so translation by  $t_{\nu(*)}$  does not change the truth of any formula  $\varphi_{p,i}$ . It follows that

$$\models \psi(a_\nu; t_k - c, t_k + c) \iff \nu(*) = k$$

and

$$\models \varphi_{p,i}(a_\nu; p^i a_{p,i}^k) \iff \nu(p,i) = k.$$

We have therefore constructed an ICT-pattern of depth

$$1 + |\mathcal{I}| = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

as required. If the sum is infinite, the ICT-pattern has depth  $\aleph_0$ . The NIP upper bound recalled in the proof of Theorem 1.4 gives  $\text{dp-rk}(G) \leq \aleph_0$ , so equality holds in this case.  $\blacksquare$

## References

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